

Remarks on: "When Hopf algebras are Frobenius algebras" by Bodo Pareigis

We work in a braided monoidal category $\mathcal{V}$.

A monoid (or "algebra") in $\mathcal{V}$ is an object H equipped with $\mu: H \otimes H \rightarrow H$ and $\eta: I \rightarrow H$, depicted by Υ and η , satisfying

$$\Upsilon = \Upsilon, \quad \Upsilon = | = \Upsilon.$$

A comonoid (or "coalgebra") in $\mathcal{V}$ is an object H equipped with $\delta: H \rightarrow H \otimes H$ and $\varepsilon: H \rightarrow I$, depicted by λ and ε , satisfying

$$\lambda = \lambda, \quad \varepsilon = | = \varepsilon.$$

A bimonoid (or "bialgebra") in $\mathcal{V}$ is an object H equipped with both a monoid and comonoid structure satisfying

$$\Upsilon = \Upsilon, \quad \Upsilon = ||, \quad \lambda = ||, \quad \varepsilon = \otimes.$$

A Hopf monoid is a bimonoid H equipped with $\nu: H \rightarrow H$, called an antipode, satisfying

$$\nu = | = \nu.$$

Proposition 1 The antipode is an antimorphism of both the monoid and comonoid structures; that is,

$$\begin{array}{c} \text{Y-shape with circle} \\ \downarrow \\ \text{V-shape with circle} \end{array} = \begin{array}{c} \text{V-shape with circle} \\ \downarrow \\ \text{Y-shape with circle} \end{array}, \quad \begin{array}{c} \text{V-shape with circle} \\ \downarrow \\ \text{Y-shape with circle} \end{array} = \begin{array}{c} \text{Y-shape with circle} \\ \downarrow \\ \text{V-shape with circle} \end{array}$$

Proof For the first equation, we look in the monoid $\mathcal{V}(H \otimes H, H)$ under convolution multiplication. From the calculation

$$\begin{array}{c} \text{Complex diagram} \\ \downarrow \\ \text{Simplification 1} \\ \downarrow \\ \text{Simplification 2} \\ \downarrow \\ \text{Simplification 3} \\ \downarrow \\ \text{Simplification 4} \end{array}$$

we see that the left side of the ^{first} equation is a left inverse for μ . From the calculation

$$\begin{array}{c} \text{Complex diagram} \\ \downarrow \\ \text{Simplification 1} \\ \downarrow \\ \text{Simplification 2} \\ \downarrow \\ \text{Simplification 3} \\ \downarrow \\ \text{Simplification 4} \end{array}$$

we see that the right side of the first equation is a right inverse for μ under convolution. The first equation follows. The second equation is dual. The third and fourth equations are exercises. $\square$

(see [Book; Proposition 9.1 (b)] or [JS; SLNMI488 Proposition 9.1])

Write $\mathcal{V}^H$ for the category of left H -modules; that is, objects A of $\mathcal{V}$ equipped with $\mu: H \otimes A \rightarrow A$, denoted by $\begin{matrix} H & & A \\ & \searrow & \swarrow \\ & A & \end{matrix}$, satisfying

$$\begin{matrix} H & H & A \\ & \searrow & \swarrow \\ & A & \end{matrix} = \begin{matrix} H & H & A \\ & \searrow & \swarrow \\ & A & \end{matrix}, \quad \begin{matrix} H & & A \\ & \searrow & \swarrow \\ & A & \end{matrix} = \begin{matrix} A \end{matrix}$$

The morphisms $f: A \rightarrow B$ preserve the action:

$$\begin{matrix} H & & A \\ & \searrow & \swarrow \\ & A & \end{matrix} \xrightarrow{f} \begin{matrix} H & & B \\ & \searrow & \swarrow \\ & B & \end{matrix} = \begin{matrix} H & & A \\ & \searrow & \swarrow \\ & B & \end{matrix}$$

There is a monoidal structure on $\mathcal{V}^H$ preserved by the underlying functor $\mathcal{V}^H \rightarrow \mathcal{V}$:

$$H \otimes A \otimes B \xrightarrow{\delta \otimes 1 \otimes 1} H \otimes H \otimes A \otimes B \xrightarrow{1 \otimes c \otimes 1} H \otimes A \otimes H \otimes B \xrightarrow{\mu \otimes \mu} A \otimes B$$

$$\begin{matrix} H & & A & & B \\ & \searrow & \swarrow & & \\ & H & & & \\ & \searrow & \swarrow & & \\ & A & & & B \end{matrix}$$

Clearly H is an object of $\mathcal{V}^H$ using its own μ . Yet, it is also a comonoid in $\mathcal{V}^H$ when equipped with δ and ϵ . To see this, we need to know that δ and ϵ are left H -module morphisms; that is,

$$\begin{array}{ccc} H \otimes H & \xrightarrow{1 \otimes \delta} & H \otimes H \otimes H \\ \mu \downarrow & & \downarrow \delta \otimes 1 \otimes 1 \\ & & H \otimes H \otimes H \otimes H \\ & & \downarrow 1 \otimes c \otimes 1 \\ & & H \otimes H \otimes H \otimes H \\ & & \downarrow \mu \otimes \mu \\ H & \xrightarrow{\delta} & H \otimes H \end{array} \quad \begin{array}{ccc} H \otimes H & \xrightarrow{1 \otimes \epsilon} & H \\ \mu \downarrow & & \downarrow \epsilon \\ H & \xrightarrow{\epsilon} & I \end{array}$$

which are bimonoid axioms.

~~Definition~~ A left H -bimodule M is an object of $\mathcal{V}^H$ equipped with a left H -comodule structure. So we have an action $\mu: H \otimes M \rightarrow M$ and a coaction $\delta: M \rightarrow H \otimes M$ such that $\delta: M \rightarrow H \otimes M$ is a morphism of $\mathcal{V}^H$; this last means

Diagram illustrating the reduction of a 3x3 grid of H and M nodes to a single M node. The left side shows a 3x3 grid where the top and bottom rows are H and the middle row is M. The right side shows a single M node.

⑤

$$P(H \otimes X) \cong X \quad \text{and} \quad Q(H \otimes X) \cong X.$$

Proposition 2 For any bimonoid H , the functor

$$H \otimes - : \mathcal{V} \longrightarrow \text{Bimod}^H$$

is fully faithful with $Q \dashv H \otimes - \dashv P$.

Proof From the equalizer for $P(M)$, we see that morphisms $X \rightarrow P(M)$ are in natural bijection with morphisms $f: X \rightarrow M$ such that $(\gamma \otimes 1) \circ f = \delta \circ f$. By the freeness of $H \otimes X$, morphisms $f: X \rightarrow M$ are in natural bijection with left H -module morphisms $\hat{f}: H \otimes X \rightarrow M$ via the equations

$$\hat{f} = \begin{array}{c} \diagup \\ \textcircled{f} \\ \diagdown \end{array}, \quad f = \begin{array}{c} \textcircled{f} \\ \diagup \quad \diagdown \\ \text{---} \end{array}.$$

The following calculations show that f satisfies $(\gamma \otimes 1) \circ f = \delta \circ f$ iff $\hat{f}$ is a left H -comodule morphism.

$$\begin{array}{c} \begin{array}{c} \diagup \\ \textcircled{f} \\ \diagdown \end{array} = \begin{array}{c} \text{---} \\ \diagup \quad \diagdown \\ \textcircled{f} \end{array} = \begin{array}{c} \text{---} \\ \diagup \quad \diagdown \\ \textcircled{f} \end{array} = \begin{array}{c} \text{---} \\ \diagup \quad \diagdown \\ \textcircled{f} \end{array}, \\ \begin{array}{c} \textcircled{f} \\ \diagup \quad \diagdown \\ \text{---} \end{array} = \begin{array}{c} \textcircled{f} \\ \diagup \quad \diagdown \\ \text{---} \end{array} = \begin{array}{c} \textcircled{f} \\ \diagup \quad \diagdown \\ \text{---} \end{array} = \begin{array}{c} \textcircled{f} \\ \diagup \quad \diagdown \\ \text{---} \end{array} = \begin{array}{c} \textcircled{f} \\ \diagup \quad \diagdown \\ \text{---} \end{array} \end{array}$$

So we have our natural bijection

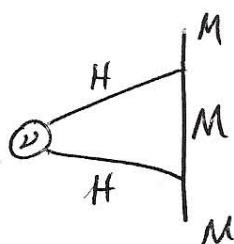
$$\mathcal{V}(X, P(M)) \cong \text{Bimod}^H(H \otimes X, M).$$

So $H \otimes - \dashv P$. Dually, $Q \dashv H \otimes -$.

The isomorphisms $X \cong P(H \otimes X)$ and $Q(H \otimes X) \cong X$ ⑥ are the components of the unit and counit for these adjunctions, respectively. So $H \otimes -$ is fully faithful. $\square$

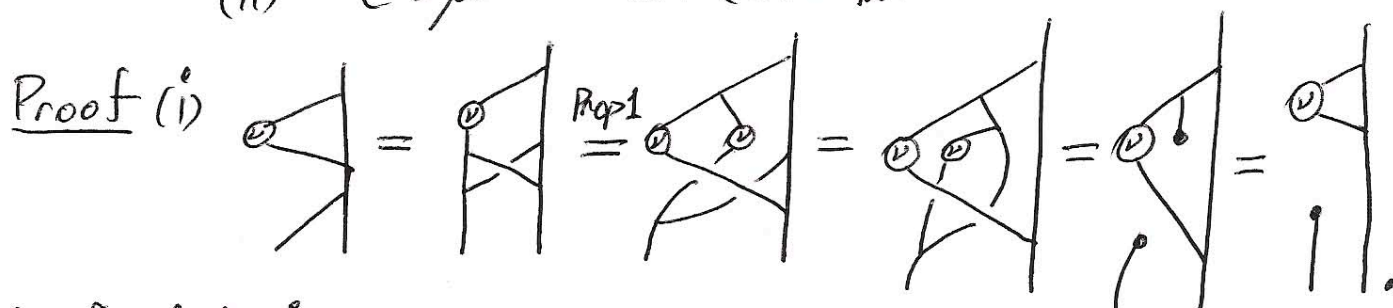
If H is a Hopf bimonoid, for each left H -bimodule M , we have the endomorphism $e : M \rightarrow M$ defined to be the composite

$$M \xrightarrow{\delta} H \otimes M \xrightarrow{\nu \otimes 1} H \otimes M \xrightarrow{\mu} M.$$



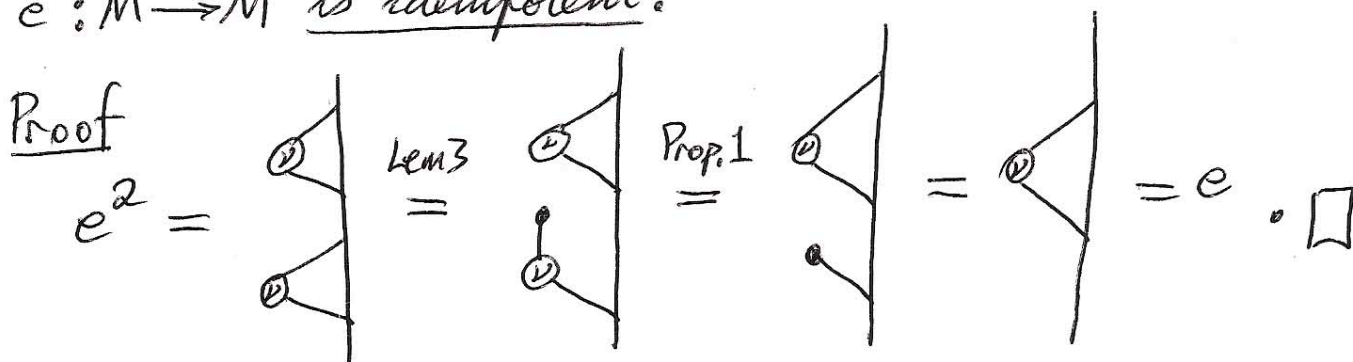
Lemma 3 (i) $\delta \circ e = (\gamma \otimes 1_M) \circ e$

(ii) $e \circ \mu = e \circ (\varepsilon \otimes 1_M)$



(ii) Dual to (i). $\square$

Proposition 4 For each left H -bimodule M , the endomorphism $e : M \rightarrow M$ is idempotent.



⑦

(i)

(ii)

Consequently, an equalizer for the pair in (i) and a coequalizer for the pair in (ii) are both obtained by splitting the idempotent $e = \mu \circ (\nu \otimes 1) \circ \delta$; the resultant equalizers and coequalizers are absolute.

Proof $\mu \circ (\nu \otimes 1) \circ (\gamma \otimes 1) =$  $=$  $=$  $=$  $= 1_M$, and

$$(1 \otimes 1) \circ \mu \circ (\nu \otimes 1) \circ \delta = (1 \otimes 1) \circ \delta \stackrel{\text{Lemma 3}}{=} \delta \circ e = \delta \circ \mu \circ (\nu \otimes 1) \circ \delta;$$

so (i) holds; (ii) is dual. The rest is standard. $\square$

Proposition 6 If H is a Hopf bimonoid then

$$H^0 : \mathcal{V} \rightarrow \text{Bimod}^H$$

is an equivalence of categories.

Proof By Proposition 5 we have the splitting

$$\begin{array}{ccc} M & \xrightarrow{e} & M \\ & \searrow g & \nearrow \pi_2 \\ & Q(M) = P(M) & \end{array}$$

$$\sigma_2 = 1_{P(M)}$$

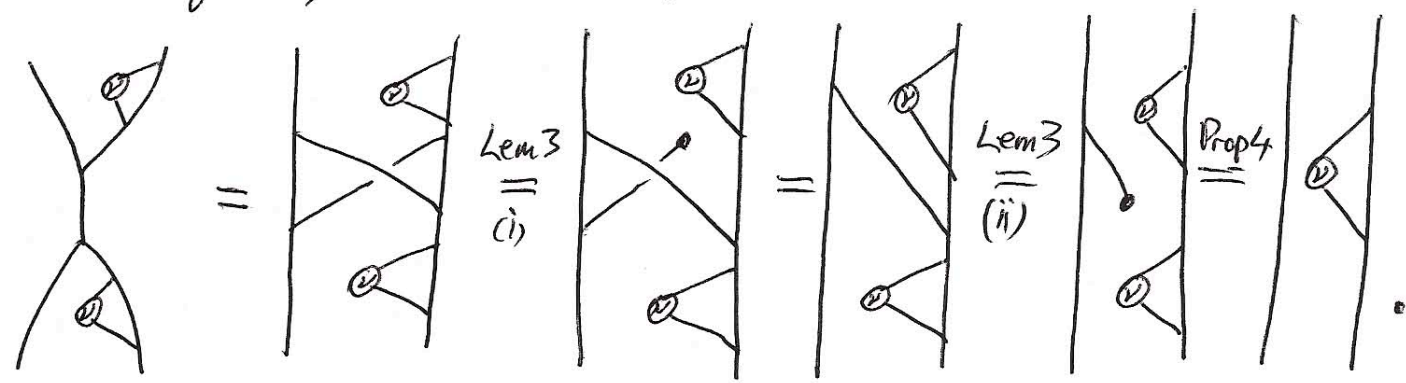
Therefore, the idempotent $1 \otimes e : H \otimes M \rightarrow H \otimes M$ has the splitting

$$\begin{array}{ccc} H \otimes M & \xrightarrow{1 \otimes e} & H \otimes M \\ & \searrow 1 \otimes \sigma & \nearrow 1 \otimes \tau \\ & H \otimes P(M) & \end{array}$$

We shall show that we have another splitting of $1 \otimes e$:

$$\begin{array}{ccc} H \otimes M & \xrightarrow{1 \otimes e} & H \otimes M \\ & \searrow 1 \otimes e & \nearrow 1 \otimes e \\ & H \otimes M & \\ & \mu \searrow & \nearrow \delta \\ & M & \end{array}$$

First of all, the last diagram commutes since

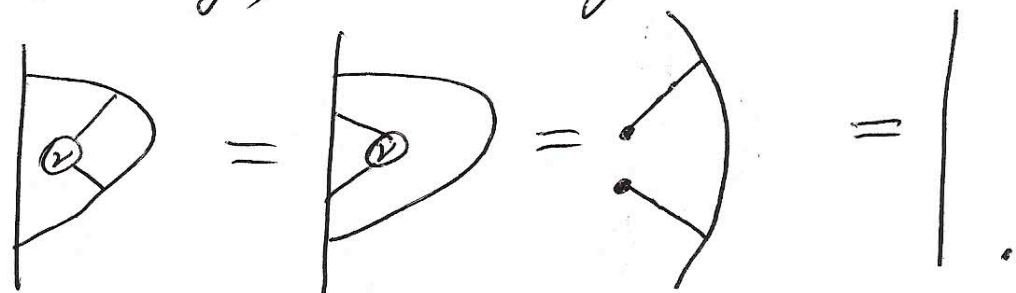


Yet also, the composite

$$M \xrightarrow{\delta} H \otimes M \xrightarrow{1 \otimes e} H \otimes M \xrightarrow{1 \otimes e} H \otimes M \xrightarrow{\mu} M$$

$\xrightarrow{1 \otimes e}$

is the identity, as shown by the calculation



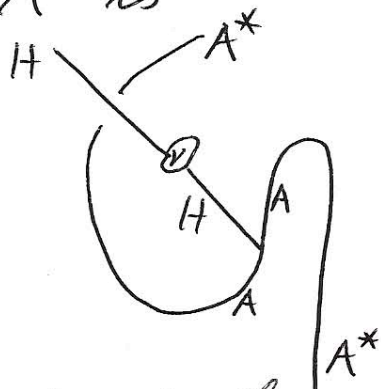
Finally notice that the morphisms μ , δ and those of the form $1_H \otimes f$ are all in Bimod^H .

(9)

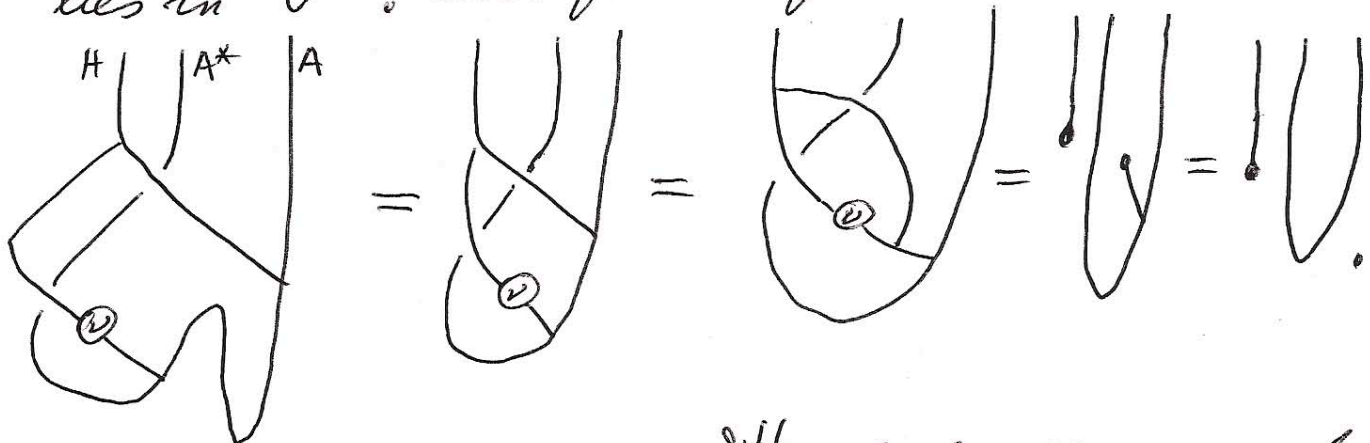
$$\begin{array}{ccccc}
 & & H \otimes P(M) & & \\
 & \nearrow 1 \otimes \sigma & & \nwarrow 1 \otimes \tau & \\
 H \otimes M & & & & H \otimes M \\
 \downarrow 1 \otimes e & & \uparrow \cong & & \uparrow 1 \otimes e \\
 H \otimes M & & M & \xrightarrow{\delta} & H \otimes M \\
 & \searrow \mu & & &
 \end{array}$$

□

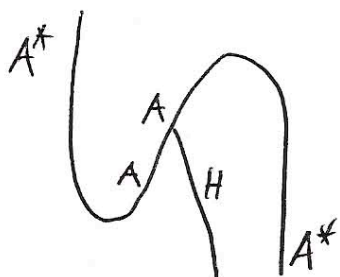
Recall from [Book, Example 12.6], an object $A \in \mathcal{V}^H$ has a left dual if and only if the underlying object $A \in \mathcal{V}$ has a left dual A^* ; the left H -module structure on A^* is



The main point is that the counit $A^* \otimes A \rightarrow I$ lies in $\mathcal{V}^H$; this follows from the calculation

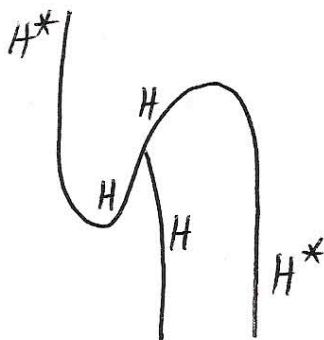
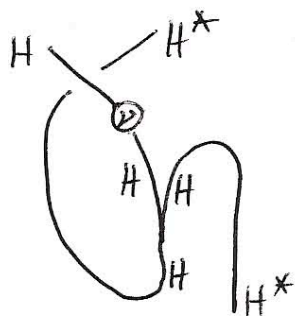


Consequently, working in $\mathcal{V}^H$, the left H -comodule structures $\delta : A^* \rightarrow H \otimes A^*$ on A^* are in bijection with the right H -comodule structures $\delta_r : A \rightarrow A \otimes H$ on A .



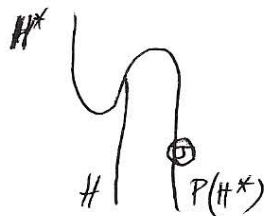
Assume H has a dual H^* in $\mathcal{V}$. Certainly then H^* is a left dual for H in $\mathcal{V}^H$ and we have the right H -comodule structure $\delta: H \rightarrow H \otimes H$. It follows that:

Proposition 7 If H has a dual H^* in $\mathcal{V}$ then H^* is a left H -bimodule via the following action and coaction.



Corollary 8 There is a canonical isomorphism of left H -bimodules

$$H^* \cong H \otimes P(H^*),$$



Example ($\mathcal{V} = \underline{\text{Vect}}_k$) To say H has a dual means it is finite dimensional. Moreover, $\dim H^* = \dim H$. So Corollary 8 implies that $\dim P(H^*) = 1$. Consequently we have an isomorphism $H^* \cong H$ of left H -modules. So H is a Frobenius algebra. $\square$